

sensors such as cameras and input sensors, which will help it make its own decisions. It will be able to do much more than what it can do with the existing sensors. Right now, it has internal sensors, which means that the feet move the way we program them to. This is the lowest level feedback."

So does this mean that the Terminator will have six legs and look like a spider, instead of Mr America? Or to be more precise, will we be able to make a mobile robot that takes stock of the obstacles in its course and maps out a path to perform its task, or is pre-programmed? "It's both actually," explains Kurien. "The Natraj is fully pre-programmed. The next step is to make robots that work on simple operator commands such as 'go from this point to that point'. While currently we spoon-feed a robot on what has to be done at each point, by then the robot will be equipped to encounter the obstacles placed in its path because they will be fed into the memory and it will react as instructed. This will require more sensors, the ability to recognise the senses, classify them and to make

intelligent use of them."

So are robo sapiens just being put to use where no good homo sapiens should go? Not really. Other jobs require an intelligent and observant metal head with a scientific background. The National Institute of Oceanography, Goa, is developing Maya, an Autonomous Underwater Vehicle (AUV) which has a bright future. Elgar de Sa, a scientist with the institute, works on the Maya. "The visions we have is several Mayas prodding the ocean floor, connecting to each other and making discoveries. We're interested more in discovery—we want to know what's happening under the ocean. It's so vast, we can't know enough using just ships," he says.

AUVs are being built in other parts of the world such as Europe and the United States. However, their main focus remains surveying the ocean. The Maya is a low-cost, small, but intelligent AUV. "In place



Prof Isaac Kurien,
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// The intelligence of a robot is measured by how autonomous it is //

right now are a few systems that are huge and require sophisticated launch capabilities and sophisticated sensors. We are not looking at these systems. What we're looking at is small AUVs that are very intelligent and highly manoeuvrable. Their low cost makes them affordable to be replicated in the future," explains de Sa. The Maya is approximately 1.5 metres in length, 1.3 metres in diameter and runs using a single motor. It can dive to a depth of 300 metres and has a range of about 30 km. The sensors used will have to be state-of-the-art and hence will have to be imported, but a lot of the hardware

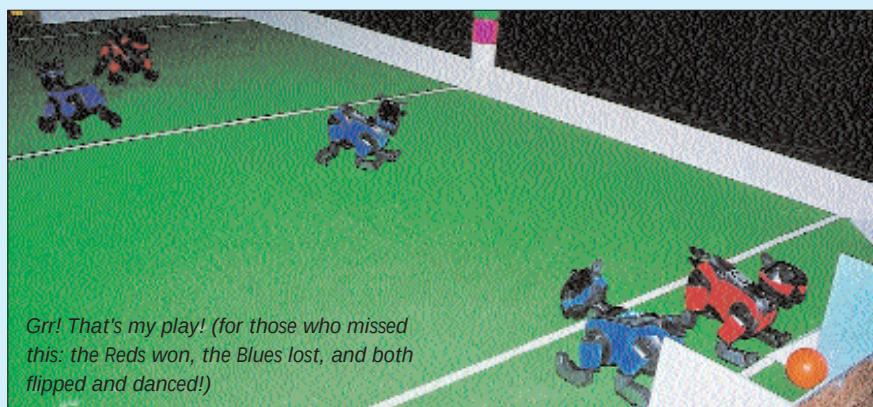
Yantriki

Yantriki is the annual robotics festival that is organised by IIT Mumbai. The brain-child of C. Amarnath, the Dean of Students Affairs, Yantriki has built up a legendary reputation of attracting the best amateur teams in the country to pit their robotic skills against each other. This year was no exception; from January 31 to February 2, teams from all over the country jumped in to take a chance at winning the top honours. Yantriki also coincides with Techfest, IIT Mumbai's annual technology festival, and the resulting carnival makes this the most fun technology event.

The competition is made up of three



WOOF! Who let the dogs out?!



Grr! That's my play! (for those who missed this: the Reds won, the Blues lost, and both flipped and danced!)

rounds, with increasing levels of complexity and automation. At Level 1, the competition involved a game board with a folded napkin in the centre. Two robots come in from either end, cross an obstacle and place

themselves near the napkin, pick it up, and scoot right back to their home. The fastest one to do this is the winner. At Level 2, the bots fight it out over table tennis. Each bot gets 10 seconds to pick up the ball and throw it back across the net. Lose out on the time frame and the opposing bot wins a point. In Level 3, the bots enter a maze and negotiate their

way out. Sounds easy? Hold on then. Here the mouse-bot is entirely autonomous—no humans can control or intervene to help out the mouse!

The bots came with tiny electrical motors, belts, wires, toggle switches, tape and aluminium foil and just about anything that could be mounted on it; one bot, for example, used a fan to whisk up the ping pong ball in the Level 2 competition. The aggressive little bots came with names such as The Titans, Sharp Shooters, Jaws, Snatcher and Balbeer Pasha. In the ensuing war, bots did break down, and there was help at hand with IIT Mumbai's Robotics Lab providing a full-fledged emergency room.

And then came the final competition